ActInf GuestStream 114.1 ~ How do inner screens enable imaginative expe

GuestStream | How do inner screens enable imaginative experience? | 1:39:47

Hello, welcome everyone.

It is July 17th, 2025.

We're in Active Inference Guest Stream 114.1 with several of the authors of

How Do Inner Screens Enable Imaginative Experience?

Applying the Free Energy Principle Directly to the Study of Conscious Experience.

Thank you to the authors who have joined into Koya for precipitating this stream.

So, without further inner screen stream ado, Chris, please, however you'd like to take it,

and I'll scroll to wherever you need.

Okay, thank you, Daniel.

Thank you, Adam and Maxwell, for being here.

And thanks to our other co-authors who are not here for whatever reason,

one of which is because we changed the schedule on this presentation by an hour at the very last moment.

So, we just wanted to talk a little bit about this paper and then open it up to questions.

So, it's good that you have some questions prepared.

So, what's the basic motivation here?

The basic motivation is simple.

If one looks at any kind of standard Active Inference picture of what's going on,

in the middle of that picture is a Markov blanket.

And, in fact, we could look at figure one because that illustrates this.

And this is all very familiar.

One has some system that's interacting with some environment.

And the interaction is mediated by a boundary.

And the boundary is what keeps the states of the system conditionally independent from the states of the environment.

It's called a Markov blanket.

In the quantum formulation of the FEP, it's a holographic screen.

They have exactly the same function.

They encode information flowing from the environment to the system.

That's called sensation.

And they encode information that's flowing from the system to the environment.

That's called action.

Of course, that's what it's called from the system's point of view.

From the environment's point of view, the system's actions are its sensations.

And the environment's actions are the system's sensations.

So, it's all a completely symmetric system with this Markov blanket boundary in between that encodes the information exchange from one to another.

So, now in this standard picture, if we ask, what can the system possibly be conscious of?

There's a fairly obvious answer.

It can be conscious of its sensations and maybe conscious of its actions,

because all that's data that's written on its Markov blanket.

And then, in some vague sense, maybe it's conscious of something that's going on inside it.

And that's where the problem is.

What does it mean to say that a system is conscious of internal states or internally generated experience?

And in the most straightforward reading of the FEP, that simply doesn't happen.

All that's going on internally is some dynamical system doing its thing, some bunch of physics,

some vector floating around in a state space.

And we can make this very precise in the quantum formulation.

And the quantum formulation, what's going on inside the system, may just be a unitary evolution of some pure state in some almost isolated Hilbert space.

And so, on the inside of the system, at least in the quantum formulation,

it's very easy to stipulate that there's no classical information at all.

Nothing that could be experienced.

No definite states that aren't big superpositions of all the possible values of all the possible degrees of freedom.

So, at least in that way of thinking or in that formulation,

if there are no boundaries inside the system that function as Markov blankets,

there's no possibility of inner experience.

And if there are boundaries in the inside of the system that function as Markov blankets,

then the experiences encoded on those boundaries

are the experiences of the systems that are bounded by those boundaries,

i.e. of parts of the system, not the whole system.

So, in this way of thinking about the FEP,

which is, again, is kind of the most straightforward possible way of thinking about it,

what a system experiences is only the data that's written on its Markov blanket,

either by its environment or by itself.

So, the motivating question for this paper is,

how do we account for imaginative experience in that kind of model?

And the answer is pretty obvious.

We've got to invoke some sort of inner boundary or inner Markov blanket or inner screen

that mediates this experience.

And we've got to somehow see the agent of interest as being inside that boundary.

So, that's what we do in this paper.

So, if you, Daniel, would please fast forward to figure three.

There it is.

I don't know whether it's possible to make that a little bit bigger on the screen.

But, yeah, that's better.

This is kind of a stripped-down picture of an active inference agent

that's complicated enough to have several perception-action loops,

which we've labeled QRF for quantum reference frame,

which is a representation of a perception-action loop in the quantum formulation.

And what controls these perception-action loops,

specifically by allocating attention to them,

and we think of attention in, again, the simplest possible way

as allocating energy to allow the things to actually function,

what controls all that is some sort of executive system,

which is a metacontroller or a metaprocessor

in kind of software language or software architecture language.

And since these perception-action loops

are exchanging information with this metacontroller,

there must be a boundary inside.

And that boundary defines what the metacontroller experiences.

What the metacontroller experiences is the activity state

of the perception-action loops.

That's what it's trying to control.

And it also experiences whatever output they give it.

So they might identify some object and tell the metacontroller,

oh, that's a table, or that's a conspecific,

or that's a predator, or that's something else.

And the system as a whole experiences what's written on its boundary,

which is labeled B here, by its environment.

So we have two different kinds of experience going on now.

We have an experience on an outer screen,

which is information encoded by the environment

or information encoded by the system's actions,

and information on this inner screen called BMC,

which is just the experience of the metacontroller.

But it's also the experience of the perception-action loops

that they get from the metacontroller.

So they can experience the data that the metacontroller

writes on their boundaries.

And these sort of black things that look like valves at the top

are meant to indicate the flow of control,
i.e. the flow of energy, resources, memory, precision,
whatever you want to call it,
from some resource pool
to these perception-action loops or QRFs.
So this simple architecture does not encode
or support any kind of perceptual experience.
So it doesn't support visual imagination
or inner speech or anything like that.
And we can claim that based on two or three decades now
of cognitive neuroscience,
which indicate that imagination,
at least in humans and presumably in mammals
and probably in vertebrates,
employs the same kind of input processing
that perception does,
up to a few details.
So when you have some sort of imaginative visual experience
or when you engage in inner speech,
your fairly low-level perceptual processes are engaged.
And this is fairly well documented.
So imagination is not just something that,
for example,
the prefrontal cortex does
or the executive network does.
It involves the sensory systems.
So to make a model of imagination
in this active inference framework
that's realistic for the animals that we know about,
we need to engage these perceptual modules.
So the solution to doing that
is actually pretty straightforward.
And if you'll scroll down to figure four,
please, Daniel.
What we do is just add another boundary.
And this boundary is labeled BN in this picture.
And we make the stipulation
that BN can and typically does encode

all of the information that's on the boundary
that's labeled here B out,
which is the Markov blanket around the whole system.
And that BN is what the system experiences.
And this picture is complicated
by issues of control.
Not only does the system,
the executive system or the metacontroller,
need to manage the activity
of the perception action modules,
these QRFs,
it needs to manage
the flow of information
between B out,
the boundary that the environment
can write directly on,
and B in.
And most of the time,
it's going to leave those channels open.
But it needs to be able to close them,
for example, by closing the eyes
or stopping up the ears
or something like that.
And then it can allow
these perception action modules
to see information
that either themselves
or other perception action modules
have written on this inner boundary.
So if you look at this picture
and think about popular architectures
like the global workspace,
then clearly what we've done here
is just distribute the global workspace
between the metaprocessors boundary
and the inner screen boundary.
And we've also indicated
by the red horizontal arrows here

that sometimes the metacontroller
may not actually be possible
to completely suppress a channel
from the external environment.
So even when you're sleeping
and dreaming, for example,
a loud enough noise
can typically wake you up.
And if you're driving on the highway
and sort of idly reminiscing
about something,
events on the highway
can rapidly bring you to your senses
or else you die.
So there's heavy evolutionary pressure
to keep these channels
openable by the external environment.
And in mammals, at least,
that's implemented
by the ventral attention system,
which can override
the dorsal attention system
and say,
no, pay attention,
no kidding,
that thing in front of you
is a tiger.
So that's what we've done
in this paper architecturally.
It's kind of followed
the standard active inference architecture
to one logical conclusion
about experiencing content
that's created inside the system
as if it was content
created outside the system.
And the remainder of the paper
talks about

what we know
about this capability,
the capability of imagination.
And the short answer is
we know a lot more about it
than we did, say, 20 years ago,
but we know a lot less about it
than we'd like to.
So we know very little, for example,
about its phylogenetic distribution.
We can look at
lots of kinds of animals,
including birds and octopi
and certainly mammals,
and infer that some of their behavior
is playing.
And some of that play behavior,
at least,
may be driven by internal imagination.
We don't know.
We can look at human variation
and see a lot of differences
in human capabilities
to have imaginative experience.
So, for example,
I'm pretty incapable
of having visual imaginative experience,
aside from diagrams.
And so are a lot of other people.
We know that imaginative experiences
are involved in various ways
in pathology.
So, for example,
we know that
one symptom of
some sorts of schizophrenia
involves uncontrolled
imaginative experience

or failures

of what's called
the reality monitoring system,
confusion between
internally generated experience
and externally generated experience.
So, a lot turns
on how imagination
is implemented.
And so it's important
for the Active Inference Project
to have an implementation
of imaginative experience
that's fully consistent
with all of the other
architectural assumptions
of Active Inference.
So that's what we're trying
to provide here.
And we've hopefully provided
a very simple,
bare-bones,
kind of minimal assumption model
of what's required
for this kind of phenomenology.
And hopefully the model
is detailed enough
to be able to
provoke
or at least encourage
various further experimentation
into
exactly
how imaginative experience
works
in the,
at this point,

only system
where we really have
access to
reports of imaginative experience
even though they may not
be very reliable
and that is us.
So,
that's kind of a summary
of what we're doing
in this paper.
And I'm happy
to
answer questions
or if they're not
questions for me,
defer them to my colleagues
who know much more
about various aspects
of this topic
than I do.
Well, Chris,
if I could just,
if I could prompt you
in a specific direction,
I think one,
one nice thing
to maybe discuss,
that was a great presentation
of the work,
by the way.
Thank you for being
so, you know,
thorough and comprehensive.
I was wondering
if you had a few words
maybe to add
on the Neo-Cartesian angle.

So,
I think there's a sense
in which
the work that we've done here
kind of vindicates
something like
a naturalization
of the homunculus.
It certainly makes
a lot of these ideas
a lot less
silly sounding.
Do you have a view there?
Well,
I would say
it,
it only vindicates
in the same way
that the,
the global workspace
model does.
The idea
of an,
of an inner screen,
an inner
locus,
if you will,
of classical information
in what could be
an otherwise quantum system
or,
or an otherwise
boundaryless,
internal boundaryless system.
It says that
to explain
certain sorts
of phenomenology

you need to have
some such construct.
Where,
where it differs
from
the
classical
homuncular picture
which may of course
be a parody
is that
the systems
that it can
experience
what's on the
inner screen
are
in a,
in a very real sense
smaller than
the entire system.
And one way
in which they're
smaller
is that there
has to be
stuff
between the
inner screen
and the outer screen.
and
that stuff
architecturally
is not
experienceable
by
the system
inside the

inner screen
in this picture.
The other way
that it differs
is that
the
executive
system
and
the
executive
system
plus
the
perception
action
loops
have
very
different
experiences.
And
in a lot
of
kind of
I'll
call it
folk
psychology
the
experiencing
agent
is
more or
less
equated
with
its

metacognitive
system.
So
if I
say
I'm
currently
experiencing
X
that's
by definition
a report
from my
metacognitive
system.
And
because it
has to be
looking at
what's
going on
inside
my
cognitive
system,
my total
cognitive
system.
And I
think one
implication of
this kind
of work
is that
those reports
are always
unreliable
because the

metacognitive
system is
always working
with a
very
stripped
down,
coarse
grained,
abstracted,
reportable
experience.
So
what I
say I'm
experiencing
is actually
not a very
good guide
to what I'm
actually
experiencing.
And a
corollary of
that is
what I can
remember
experiencing
is not a
terribly good
guide to
what I'm
actually
experiencing.
So it
does make
those challenges
to kind

of folk
psychological
notions of
kind of
inner life.
things.
Yeah,
thanks for
that.
That was
very
interesting.
So I
know,
Adam,
yeah,
you unmuted
that.
I was
hoping you
also had
something to
add to
that.
I know
neocartesianism
has been one
of the
things that
like,
I think I
got the,
you know,
I was first
introduced to
this idea
by you and
Sam Vaseer

about at the
same time.
so hopefully
this also
sparks,
you know,
interest,
maybe even
joy.
Interest,
joy,
and trepidation.
Like,
I guess I can
blame you for
partially like
not giving up
on the idea
because it's
so unpopular
because it's
just like,
you know,
Dennett was
like so
like clear
and firm
on this point
and so many
people joined
him on like
Descartes'
errors.
And I mean,
clearly there are
problems there
and I can't
deny what

like,
you know,
with what
Chris is saying
about like,
we can't
just like
blindly trust
like folk
psychology
or our
introspective
access.
And we know
there has to
be distortions.
But I guess
one thing
I was wondering
about is
whether some
quasi-Cartesian
intuitions
might actually
reflect both
joints in
phenomenological
space and
potentially
guides to
how they're
realized.
Like in terms
of what sorts
of implementations
or computational
objects they
might be.

And so
like this
work from
Chris,
you know,
it comes
from a very
different
background,
even though
like I'm
also,
you know,
inspired by
the Franergy
principle.
Chris is
coming from
like the
fundamental
physics and
like,
you know,
like how
quantum systems
must have
like a
singular
interface
where locality
is created
in order to
achieve,
I'm going to
butcher what I'm
saying,
but like,
like the way

you're approaching
this is very
different for
me,
but very
convergent.
And so
then this
idea of
like an
inner screen,
that's
something I
was suggesting
from my
reading of
both like
cognitive
neuroscience,
like neuroimaging
literatures,
and also
attempts at
trying to
make cross
mappings
between
machine learning
architectures
and
understandings of
neural function
and how they
might,
you might
interpret the
brain as a
kind of hybrid

machine learning
architecture
and generative
model.
But with
this,
so in some
of the work,
some of the
explored with
Tim Verbalen
in the context
of robotics,
the idea is
like interpreting
the brain as a
cybernetic
controller for
an embodied
agent.
That's your
slam paper
with Tim,
right?
Yeah,
the slam
paper.
Yeah,
cool,
very cool.
What's cool
is like in
this work,
they have
basically one
of their
core data
structures

they use
for robot
autonomy is
a view
and a pose
network.
And so I
don't want to
over-interpret
or over-stress
the connection,
but I've
sometimes somewhat
fancifully,
but perhaps I
should just
commit to the
crime,
thought of
as almost
like a kind
of quasi-Cartesian
theater and
homunculus
and their
relation.
And the idea
is like you
have the
view of the
robot and
its pose
as being
two very
closely entangled
data structures
that are
basically almost

that are
heavily mutually
informing each
other because
what you're
seeing is
informed very
heavily about
where your
sensors are
pointed and
that also what
you're seeing gives
you a sense of
what your body
is doing.
And so these are
the two primary
data structures
and these are
also, if you
look at, I guess,
like neural realist
state and cortical
hierarchies, like
the brain is
dominated by
vision and
then touch.
and so the
idea I've
been exploring
and it
wasn't, I
keep kind of
being drawn
back to, it's
not part of the

hardcore of any
theory that if it
was falsified, it
wouldn't like
disprove things
like the overall
framework, but if
it was like shown
to be evidenced, I
think what provides
some support is
that potentially
like inner
portions of
like the
cortical hierarchy
as a kind of
shared latent
workspace could
potentially operate
by principles of
geometric deep
learning and
potentially correspond
to something like
a visuospatial
modality and
somatospatial
modality, which
are mutually
coupled and
informing.
Something like a
Cartesian theater
and a homunculus.
And the idea
being that this
being something

like an inner
screen and
observer, even
though it's not
really strictly the
sense that,
like you don't
have like homuncular
regress and that's
not like the
you have like
the clear problem
of like adducing
a fully competent
homunculus as an
explanation which
requires a homunculus
inside it.
The idea of
something like a
centralized body
like data structure
in relation to
something like a
centralized visuospatial
modality and that
these two mutually
inform each other is
part of what brings
coherence to the
sensorium.
So in Chris's work
here it's like
showing how
coupling quantum
systems you need
a kind of
singular interface

to establish
locality and
coherence.
I'm going to need
you to like
rescue me, Chris,
because I shouldn't
be speaking to
some speaking out of
school.
But at least
like shamanistically
I interpret this
as a kind of
convergence of
some quasi-Cartesian
intuitions.
Now how far you
want to go with
them, I don't want
to say like we've
like vindicated
folk psychology
and like the
points you made
about like you
have to like, you
know introspection
has to get it
wrong.
It's the whole
point of, you
know, maps have
to be simpler in
their territories,
they have to
diverge.
But I do

wonder whether
like what you've
arrived at here
could actually
provide some
vindication for
some folk
intuitions of a
quasi-Cartesian
variety of the
kind even that
Dennett for all
those years
cautioned us
against.
That's, you
know, I know
that's my
personal, one
of the ways
I'm thinking
of this, but
you know, I
know it's like
nothing you're
suggesting actually
firmly establishes
that or depends
on it.
So that's my
ramble.
I don't know
if that made
any sense.
I thought that
was a great
ramble.
Thank you,

Matt.
Yeah, I,
you know, what
Dennett and
others fundamentally
cautioned against
was kind of
regressive
homuncularism
where every
layer basically
has the same
conceptual structure
and every layer
has basically
the same
experience.
And of course
that won't
do, right?
So I think
what the,
what the
Markov
blanket
construction
I think
forces us
to do
is be
very
careful
about
saying
who has
access to
what
information

within
any
complicated,
layered,
hierarchical,
multi-component
cognitive
system.
And
because it,
any such
componentization
makes each
of the components
an agent.
and so
each of the
components
has an
environment
that they
exchange
information
with.
And so I
think what
the,
what the
FEP
encourages
is just
being very
careful
about that
and
not
assuming
that a

component
has
access
to
information
that it
can't get
from some
part of
its
environment.
So if we
think of
multiple
components
interacting
with a
shared
memory,
which is
basically
what the
GNW
is,
then
they have
access to
whatever
information is
written on
the shared
memory.
And
that's
not
kind of a
difficult thing
to think

about.
And
that
information
may be
fairly
rich,
but
what it
isn't
is the
same as
the total
experience
of any
combination
of those
systems.
Each
system
has
its own
experience
of whatever
is written
in that
shared
memory.
And
kind of
the
broadest
brush
access to
that
information
is more
or less

by definition
what we
call the
meta
controller
or the
executive
system
or something
along those
lines.
But again,
that's always
coarse-grained.
So,
yeah,
I don't
think the
two views
are
inconsistent.
active
inference
picture of
things
is just
a way
of being
precise
about it.
the

Thanks.
Koya,
do you
want to
ask one
of your

questions?
So,
yeah,
I was
thinking about
if the
metacognitive
system is
like a place
to gather
the memories
or,
yeah,
it's like
the reason
why we
have shared
memories
in the
first place
in the
way around,
like a
locus to
which the
memories of
the other
systems can
be shared
too,
or something
like that.
but,
yeah,
that's
highly
speculative.
So,

I was
thinking like
one thing
we didn't
talk about
at all
was this
like the
quantum
reference
frames.
What
we
understand
under
this,
we
skip
like the
figure
and
where
they
end.
Is
like the
coarse
grained
structure
on the
retina
or something
like this,
is this
part of
quantum
reference
frames

or are
they
starting
later in
the
hierarchy
with
concepts
like
a
gestalt
or
something
like
that.
Yeah,
so,
that would
be one
question.
How deep
the quantum
reference
frames we
are looking
here in
the
inner
screen
are.
I just
lost your
audio.
Oh,
I just
lost you.
Oh,
could

the
other
person
hear me
or is
it?
Yeah,
just
repeat the
question.
Yeah,
so,
we didn't
look at
the quantum
reference
frames.
Can you
hear me,
Chris?
Yes.
Yeah,
okay,
so,
we didn't
talk about
the quantum
reference
frames.
And,
so,
one question
would be,
what are
they in
a short
answer?
And then

another
question
related to
this is,
how deep
the
neuronal
architecture
are we
talking about
here?
So,
are we
talking about
something like
pixels or
single
receptors in
the retina
or are
we
talking
about
representations
further down
in the
hierarchy?
So,
how deep
does this
kind of
reference
frames that
are in
the
inner
screen?
Okay,

well,
yeah,
thank you.
I can
answer both
of those
at least
to some
extent.
so,
this
construct
of a
quantum
reference
frame
that we
use
is
actually
a very
old
idea
that's
been
formally
updated
in the
last
two or
three
decades
really
under
pressure
from
the
quantum

communication
and
quantum
security
communities.
So,
in classical
physics,
you have
things called
reference
frames.
For
example,
Cartesian
coordinates
are a
reference
frame.
And
these are
things that
we use.
So,
clocks,
meter
sticks,
all of
those are
reference
frames.
what does
that
mean?
Well,
it means
two
things.

A
reference
frame
is both
a
standard,
so it
defines a
kind of
unit,
for example,
the meter,
and it's
a way
of measuring.
measuring.
So,
a meter
stick
lets you
measure
things
in
meters.
So,
it's
those two
concepts
kind of
bundled
together.
And in
classical
physics,
these things
are treated
as
abstractions.

If you
read
Einstein's
work,
for example,
he's
constantly
talking about
moving
frames of
reference,
et cetera,
and observers
sitting on
moving frames
of reference
and reporting
what they
see,
and on
and on.
So,
as
quantum
information
started to
become a
discipline,
it was
recognized
early on
that this
purely
abstract way
of thinking
of reference
frames
wouldn't

do,
that the
physicality
of reference
frames actually
had to be
taken into
account.
So,
a meter
stick
is not
an abstraction,
it's an
actual object,
you know,
it's a piece
of wood
or a piece
of metal
that we
hold in
our hands
and we
use it
to interact
with the
world and
those interactions
are actually
forceful and
they require
energy and
so on.
measurement
is not
this
completely

passive
abstract
thing
where by
magic you
get some
information.
You have
to actually
kind of
whack the
world with
something,
with your
meter stick
to get some
information out
of it.
And that's
the fundamental
reason for
things like
Heisenberg's
uncertainty
principle.
So,
if you
think about
reference
frames in
this physical
way,
and then you
think about
observers as
physical systems
like us,
and you

ask,
so what's
required for
me to be
able to use
a meter
stick?
I used to
have one
around here,
but I don't
see it right
now.
So,
if I have
this object,
a meter
stick,
and I
want to
measure
length,
then I
have to
have some
concept of
what length
is,
and that
concept has
to be encoded
in my
brain.
It has to
be part of
me as a
physical
device.

And,
if I use
a clock,
I have to
have some
sense of
time.
I have to
be able to
notice that
the hands
on the
clock have
moved,
for example.
So,
I have to
have a
clock in
my head.
I have to
actually be
a clock in
order to
use a
clock somewhere
out in the
external world.
Otherwise,
I won't even
notice that
the clock is
moving.
So,
that led us
to think
about what
does it

mean to
say that
the observer
implements
quantum
reference frames,
that components
of the
observer
actually are
quantum
reference frames.
And,
we know
from neuroscience,
for example,
that the
hippocampus
encodes a
very sophisticated
3D reference
frame.
And,
we know
that some
of the
perihippocampal
areas encode
fairly sophisticated
clocks.
So,
we're not
making this
up.
It has some
connection to
neuroscience.
It's not

completely
theoretically
motivated.
So,
now we
can ask,
how do
we want
to conceptualize
these things?
And,
again,
we think of
meter sticks
and clocks
as just
objects in
the environment.
And,
they may do
something like
clocks,
they may have
to use energy,
and they may
just sit there
like meter sticks.
and they
may be
outcomes of
some complicated
physical process.
So,
for example,
if I want
to define
what up
means,

if I'm
measuring spin,
I'm generally
going to do
it with reference
to the Earth's
gravitational field.
Well,
what's that?
Right?
It has
something to do
with the
curvature of
local space
time.
But,
it defines
an external
reference frame
that I can
use.
I can use
the gravitational
field or
the magnetic
field to
think about
what up
means.
But,
that requires
that I
have some
representation,
that I
implement some
representation

that lets me
access by
interaction
that external
direction,
directionality,
either magnetic
or gravitational.
gravitational,
and we
know that we
use gravity
all the time
to move
around.
That's one
of the key
features of
our bodily
system.
We know
that birds,
for example,
use magnetism
in migrating,
that kind of
thing.
So,
again,
this is stuff
that we know
about from a
neuroscience
perspective also.
So,
to sort of
come to a
close on the

first question,
a quantum
reference frame
in this way
of thinking
is a
component
of the
internal
process of
some observer.
And you can
think of it as
a component of
the internal
process of
some organism.
And,
theoretically,
that internal
process is just
some dynamical
system or
can be
represented by
some Hamiltonian
operator or
something simple.
And,
of course,
we have no
clue about how
to write those
operators down,
but formally,
they're easy to
play with.
So,

you can think of
a quantum
reference frame
as just some
components of
a dynamical
system or
a Hamiltonian.
So,
now to the
question of
how deep
are they.
And that's
a subtle
question,
because
one can
construct
an arbitrarily
deep
hierarchy of
operators,
and if
those operators
all have
a property,
so in
quantum theory,
for example,
all measurement
operators are
Hermitian,
which just
means that
their eigenvalues
are real,
the outputs

that they give
you are real
numbers,
and so you
can,
since they're
finite resolution,
you can think
of the outputs
as binary
numbers,
bit strings.
You can
construct an
arbitrarily large
hierarchy of
these operators,
and the whole
hierarchy will
be Hermitian,
so it will
look at the
world and
give you
a real
bit string,
provided one
condition is
met,
and the
condition that
has to be
met is the
operators have
to commute.
That's what
those arrows
mean in this

diagram that's
on the
screen.
And in
classical physics,
of course,
all operators
commute,
but in
quantum theory
they don't.
So quantum
theory has the
great advantage
of giving us a
formalism that
automatically lets
us talk about
failures of
commutativity,
which means that
if I do
operation A and
then B,
I get a
different answer
than if I do
B and
then A.
And clearly
a lot of our
interaction with
the world has
that structure.
If I do one
thing and then
another thing,
the second

thing often
depends on the
result of the
first.
And it can
depend causally,
or it can
depend non-causally.
And if it
depends non-causally,
then that's
quantum
non-communativity.
Both kinds
occur all the
time.
So one
advantage of
the quantum
formalism is
that it
allows you
to deal
with
non-communativity
between
operators,
which translates
as the
inability to
define joint
probability
distributions over
sets of
variables.
But if you
can define a
joint probability

distribution over
a set of
variables,
however large
it is,
you can define
one QRF
over it.
And so you
can think of
things that,
for example,
in the
early dorsal
visual system
identify moving
edges.
Okay,
that's a
reference frame
that identifies
moving edges.
You can
identify lots
of those in
the visual
field.
Or you can
think of
something that
identifies
chairs.
Or you can
think of
something that
identifies
scenes.
or you can

think of
something that
identifies
short
histories of
different
scenes.
And as
long as
the operators
that build
up those
identifiers
commute,
you can
think of
that as
one quantum
reference frame.
so you
automatically
get a
very deeply
hierarchical
system that
nonetheless
divides,
splits into
fragments when
you have
operations that
don't commute,
when looking at
the world
changes it.
you want to
respond or
ask anything

else for you,
please.
Thanks for
the answer.
Really
interesting.
I was thinking
about what to
do with
Adam's
questions or
remarks.
should I
address them
now or
later?
Sure.
You want to
talk about it?
Your thoughts?
Should who
address what?
Adam wrote
like four
messages.
Yeah,
Adam,
if you write
something in the
chat, please
say it on the
stream.
Okay.
So,
Chris,
in your
work in
terms of

quantum
interpretations
of the
FEP,
some people
might pattern
match and
then say,
oh,
you mean
orchestrated
reduction and
quantum
computation.
But no,
it's like a
quantum
formalism
formalism for
the use
of quantum
formalism.
And it
makes me
wonder,
I reached the
limits of my
technical depth
here,
but in terms
of how far
we can go
with which
quantum
intuitions and
interpretations.
of the
equations.

So,
for instance,
when you're
thinking of
interacting
neuronal
populations,
as they're
coming in
and out of
coherence,
and they
have probabilistic
properties to
where they're
at in terms of
their degree of
coherence,
as they
form,
they couple
various types
of synchrony,
these interacting
clouds of
neurons,
is there a
sense in
which part
of the
reason you
could get
quantum
formalism
from a
neuronal
system is
because of

its probabilistic
nature,
like some
sort of
mapping between
Bayesian
brain-like
ideas,
like the
Born rule
of total
probability.
But one
of the things
I was wondering
is if there
is this
requirement
in order
to get
coherence
about multiple
observers
with multiple
reference frames
that you get
this establishment
of a singular
interface across
them with
locality and
also coherent
temporality,
but locality
at least as I
understand it.
Could this,
for instance,

like,
so is this
like requirements,
maybe something
that's also sculpted
by natural
selection or
development to
some degree,
like GenX
helps along,
but like,
could this be
part of what,
could this get
as a
precondition coherence
as like
neural populations
find where
ways of,
and the
interacting
ways to
achieve coherence
that they
establish,
the creation
of a
inner screen
with locality
as part
of this
requirement
for coherent
coupling.
And so it's
like,

again,

a ramble,

but

thinking

of quantum

derivation

of inner

screens,

it makes me

wonder about

quantum formalism

as a source

of mathematical

interpretations

of neural

functioning,

CF,

Bayesian

brains,

and born

relative

probability,

whether

interacting

neural populations

as fuzzy

blink

and

probabilistic

sets,

each with

their own

reference frame,

induce the formation

of semi-centralized

inner screens as a

couple,

do the requirement

of its creation
for coherent
coupling with its
locality and
temporality,
informational
properties,
and as a source
of,
and this also
providing a source
of phenomenal
coherence slash
binding,
as a source
of something
like a quasi-Cartesian
inner theater,
and perhaps
even a basis
for egocentric
projective geometric
visual-spatial
awareness and
sensation slash
imagination.
So that's a lot
of different things,
but I was just,
I'm wondering,
like,
you know,
I'm at the limits
of my technical
depth here of,
like,
where I should
and shouldn't

go with
drawing from
ideas from
quantum physics
and whether
what I said
is somewhat
in the ballpark
of, like,
mappings
between neuronal
functioning
and quantum
formalism.
Yeah,
I think this
is very much
worth pursuing.
I wouldn't
be surprised
if
neuronal
coherence
is not
constantly,
essentially
drawing
different
boundaries
in the
nervous system.
across which
classical
information
is forced
to flow.
One reason
for thinking

that is
that phase
coherence
is so
important
and
phase
coherence
is,
I mean,
quantum
coherence
is a measure
of a
particular
kind of
unobservable
phase
coherence.
And
phase
coherence
is something
that we
can't observe
locally,
right?
we as
outside
observers
can't
observe
phase
coherence
locally
unless we
just
happen to

look at
the right
two neurons
or the
right two
parts of
some
assemblage,
right?
We won't
notice that
this one
over here
happens to
be phase
coherent
with that
one over
there unless
we're looking
correctly.
And that's
kind of what
unobservability
means.
there's
reasonably
good
experimental
evidence
that
human
cognition
exhibits
intrinsic
contextual
or non-causal
contextuality.

the relevant
work was
done by
Edebar
Zafirov
and Andrei
Krenikov
and their
collaborators.
It's very
careful
experimental
work and
they understand
the mathematics
intimately
well.
so from
that
perspective
I mean
Krenikov
insists on
calling it
quantum-like
but I
think if
it can
only be
described
correctly
using the
quantum
formalism
you might
as well
just call
it a

quantum
phenomenon
because
quantum
theory
doesn't
have any
intrinsic
scaling
or anything
like that.
I mean
we associate
it with
the microscopic
for historical
reasons
but black
holes are
quantum objects
which are
some of the
biggest things
we know
anything about.
So I
think there's
a lot to
be done
there.
I think
probably
the
formalism
that
quantum
theory
provides

will actually
make it
much simpler
to think
about neural
behavior.
because it
does build
in these
kinds of
phenomena
like phase
coherence
that we
know
experimentally
are important
but we
lose our
ability to
think about
when we
just think
about things
like firing
rate.
that was
super interesting
I'm going to
have to drop
unfortunately
I had a
hard stop
now
but I think
Carl just
joined us
is that the

case?
yes
excellent
so it's
like a relay
race
it's good
to hear
from you
Carl
so have
a good
rest of the
conversation
everyone
thank you
very much
thank you
Maxwell
all right
thanks
Maxwell
good to
see you
all
yeah
I'm going
to have
to depart
fairly soon
but I can
at least
briefly bridge
to Carl's
part of the
conversation
if you
all want

to carry
on
what's
happened
so far
Carl
is just
a brief
description
of what
the paper
is about
and
followed
by
some
conversation
about what
quantum
reference
frames are
what
inner
screens
are
and we're
just
getting
to
thanks
to
Adam
a
discussion
of
how
coherent
activity

among
distributed
neurons
may
draw
different
effective
boundaries
and hence
construct
different
effective
Markov
blankets
within
the
system
so
you know
you know
far more
about this
than I
do
I suspect
not
but thank
you for
orientating
me
maybe
to
connect
with the
earlier
discussion
and
continue

on
Carl
how would
you see
this
in light
of the
theater
discussion
and
responses
what
unique
explanations
and
predictions
come
from
bringing
this
work
together
that
were
absent
packets
and parcel
and other
work
that had
active
inference
perspectives
on
awareness
and
consciousness
but not

the
quantum
do you
want to
just unpack
the
theater
issue
it
it
came up
a little
bit
but please
however
you want
to
summarize
and
continue
for my
benefit
just to
sort of
focus
me
what
specifically
were you
referring to
when talking
about
theater
like
your
paper
response
to the

theater
critics
oh
the
theater
sorry
I thought
you meant
theater
rhythms
oh
yes
right
it was
in a
different
world
my
apologies
let me
let me
orientate
my
response
to the
theater
critics
that was
with
Alan
Hobson
yeah
so
that was
that was
some
time ago
2016

right
yeah
well that's
nearly 10
years ago
can you
can you
remind me
what
what you
took
from that
paper
um
I'll just
summarize from
the abstract
you wrote
in brief
we fully
concur with
the theater
free
formulation
offered by
DeLega
and Dewhurst
and take the
opportunity to
explain why
and how we
use the
Cartesian
theater
metaphor
we do this
by drawing an
analogy between

consciousness
and evolution
the analogy
is used to
emphasize the
circular causality
inherent in
the free energy
principle
aka active
inference
earlier times
we conclude
with a comment
on the special
forms of active
inference that may
be associated
with self-awareness
and how they
may be
especially
informed by
dream states
and it was
in that sort
of era
with the
I think
therefore I
am
and the early
sorts of
recursive
Bayesian
models that
you were
proposing

and working
on
so
how do you
feel like
this
paper extends
and continues
in that
line
what do we
gain in
terms of
explanation
and prediction
from bringing
in
quantum
formalism
I can only
answer that
really from
a classical
perspective
but I think
a lot has
been
added to
the stories
and a lot
of nuance
has crept
into it
which is
nicely framed
by the
quantum
information

formalism
I'm trying
to cast
my mind
back a
decade
and what
was around
then
and what
is around
now
and
certainly
there's a
lot more
literature
addressing
consciousness
in its
various
guises
and in
particular
my mind
goes to
a direct
confrontation
of issues
of experience
and
phenomenology
and within
that space
the role
of
hierarchies
and

specifically
the role
of
Markov
blankets
in defining
those kinds
of hierarchies
which begs
the question
if there
are
hierarchies
in the
brain
first of
all
is there
an
irreducible
Markov
blanket
in the
sense
that
there
can
be
no
further
sparsity
of coupling
or
conditional
independences
or
dynamical
architectures

that
admit
any
further
Markov
blankets
so
this
was
one
of
the
contributions
from
the
classical
perspective
to
an
early
formulation
of
the
paper
that
Chris
has
just
taken
us
through
the
other
thing
that's
brought
to
the

table
is
if
there
are
Markov
blankets
within
a
brain
what
constitutes
the
active
states
of
any
of
these
internal
Markov
blankets
so
that
becomes
an
interesting
focus
and
the
story
that
people
like
Lars
Smith
and
company

including
people
like
Mark
Soames
would
pursue
is
that
the
way
that
you
act
upon
things
lower
in
say
a
centripetal
or
centrifugal
hierarchy
would be
primarily
through
mechanisms
like
attention
again
in the
classical
context
would be
read
as
optimising

the
precision
or the
confidence
ascribed
to
various
representations
at
lower
levels
and
that's
I
think
an
interesting
move
because
that
weds
the
psychological
interpretation
of that
gain
control
active
gain
control
to
attention
in the
psychology
literature
it
also
provides

a nice
link
to
Mark
Soames
notion
of
felt
uncertainty
remembering
that
precision
is the
complement
or the
inverse
of
uncertainty
and
that
there
are
people
who
would
regard
any
feelings
or
qualitative
experiences
would
require
there
to be
a
mechanism
in the

brain
that
would
render
what was
transparent
opaque
that would
be
attention
or would
from the
point of
view
of
simulations
be
the
gain
control
that
comes
along
with
this
kind
of
mental
action
and
I
use
the
word
mental
action
as
another

way
of
describing
the
role
of
the
process
that
would
be
implicit
in
having
active
states
on
Mark
blankets
in
the
brain
so
talking
about
a
covert
kind
of
action
and
that
covert
action
can
be
read
as

attention
or
it
can
be
read
as
sort
of
actively
selecting
those
sources
of
evidence
for
belief
updating
within
on the
internal
side
of
the
Mark
blanket
so
I
think
that's
quite a
nice
story
that
is
quite
accommodating
of

a
number
of
different
perspectives
and
different
kinds
of
formulation
but
let's
come
back
to
this
notion
of
an
irreducible
Mark
off
blanket
I
think
that
in
itself
is
an
interesting
concept
and
in
many
senses
when
I

read
Chris
talking
about
the
inner
screen
what
could
argue
and
indeed
a lot
of
people
do
argue

Chimil
being
one
of
these
people
Ruben
along
with
Chimil
being
one
of
these
people
that
there
are
a whole
host

of
inner
screens
and
I
think
the
most
recent
analogy
is
that
of
these
smart
glasses
that
through
mental
action
you
can
render
opaque
or
transparent
with
screen
after
screen
after
screen
so
what
is
the
inner
screen

and
to
my
mind
just
in
a
very
simple
minded
way
that
has
to
be
the
innermost
screen
that
surrounds
the
irreducible
Markov
blanket
and there's
some
interesting
issues
that
attend
that
well
if
this
irreducible
Markov
blanket
is

responsible
for
providing
messages
and if
we take
quantum
theoretic
formulation
then
we're
talking
about
the
holographic
screen
that
is
surrounding
the
irreducible
bulk
that
constitutes
this
irreducible
part
of the
system
and
writing
that
classical
information
to
the
active
sector

of the
holographic
screen
has to
have
consequences
and what
are those
consequences
what we've
just said
from a
sort of
neurophysiological
point of
view
it has
to engage
those
mechanisms
that
mediate
gain
control
and
immediately
you're
now drawn
to
thinking
okay
well that
basically
means
that
the
projections
of the

inner
screen
must
in some
way
do
some
kind
of
attentional
selection
beyond
the
screen
and
that's
the
sort
of
mental
action
that
I
was
referring
to
previously
so
I
think
the
notion
of
the
inner
screen
is
something

which
we
didn't
have
until
a few
years
ago
and
if you
like
to my
mind
is
a
denouement
of
arguments
that have
been
formalized
and developed
over the
past
10
years
I'm not
sure
that speaks
to
the
theater
debates
they were
more
predicated
on
if I

remember
correctly
dual
aspect
monism
that
inherits
from
the
two
kinds
of
information
geometries
that
you
get
from
a
classical
physics
perspective
where
you're
linking
the
movement
on
a
statistical
manifold
to
an
information
geometry
which
essentially
encapsulates

a belief
space
so literally
changing
your
mind
to
exactly
the
same
or
supervening
on the
same
substrate
which would
be a
neuronal
substrate
that you
could
cast
in terms
of
neural
firing
that
has
itself
an
information
geometry
read as
classical
stochastic
thermodynamics
so whenever
there's

a
space
of
probability
distributions
say of
the kind
that underwrites
statistical
or
stochastic
thermodynamics
there is a
thermodynamics
so there's a
natural link
there between
the sort
of
the
energetics
of the
brain
and
the
informational
or
representational
geometry
on two
statistical
manifolds
that
supervene
on
the
same
substrate

how
that
perhaps I
should pass
back to
Chris
now who
hasn't
gone
away
so he's
still
here
so
Chris
would
wrap up
that
thermodynamic
aspect
into
a
free
energy
thermodynamic
free energy
sector
on
the
holographic
screen
so
I think
there's
an
interesting
discussion
here

as to
whether
there is
a
dual
aspect
well
whether
there are
any
information
geometries
in the
spirit
of
the
statistical
manifolds
and if
there
are
how are
the
thermodynamic
versus
the
classical
information
structures
linked
and I'll
have a
rest
now
was that
good
enough
thank

you
very
much
great
wow
I
want to
change
directions
away from
the
theater
to
something
else
we've
on the
paper
you talked
about the
phylogeny of
imagination
and you
gave some
reasons why
the second
layer in
the screen
was implemented
and there
was the
advent of
covert action
social
situations
could be
better managed
if some

actions are
covert
and the
other thing
was the
planning
depths
I was
thinking
maybe
there's
this
cybernetic
concept of
second order
observation
to use
the inner
observation
on something
else
to see
what
one person
is using
to experience
this thing
so to
reflect
on
the
own
source
and I
think
this
idea
of this

inner
screen
could
really good
at capturing
this idea
so we
can use
on the
inner
screen
a
quantum
reference
frame
to look
at another
quantum
reference
frame
and to
look
what
quantum
reference
frame
we used
there
like
the
second
observation
this
could be
maybe
like
another
advent

for
this
kind
of
architecture
and
could
this
be
achieved
without
this
inner
screen
yeah
I
think
a very
good
question
is
how
does
the
system
represent
its own
capabilities
so
in
the
kind
of
simple
minded
picture
that
we

drew
the
metacognitive
system
or the
controller
the
attention
system
has to
know
something
about
what
the
various
perceptual
and
action
capabilities
of the
organism
were good
for
so
that
it
can
decide
what
to
spend
energy
on
what
to
spend
energy

doing
and
this
we
know
that
this
goes
way
way
down
in
phylogeny
so
we
know
that
even
bacteria
have
very
sophisticated
ways
of
for
example
swapping
in or
out
whole
metabolic
pathways
to
deal
with
different
food
sources

that
are
present
in
the
environment
and
a
key
discovery
from
the
60s
and
early
70s
was
how
this
works
how
the
food
source
itself
is
perceived
and how
the
perception
of a
food
source
can
trigger
the
expression
of

whole
families
of
genes
that
then
encode
enzymes
for
metabolizing
and using
that
food
source
so
you
can
think
of
that
in
terms
of
meta
control
you
can
think
of
that
as
a
system
saying
oh
here's
something
in the

environment
that
tells
me
that
I
should
turn
off
a
whole
bunch
of
my
information
processing
and
turn
on
a
whole
other
bunch
of
my
information
processing
so
in a
sense
I
should
switch
what
I'm
paying
attention
to

and
switch
what
I'm
doing
about
it
I
should
turn
on
whole
different
perception
and action
capabilities
as
units
so
we
know
that
that
kind
of
capability
is
extremely
deep
in
phylogeny
all
organisms
have
that
kind
of
capability

so
when
we
get
up
to
these
very
complex
organisms
that
actually
have
brains
we
can
ask
what's
the
analog
of
that
at
the
level
of
for
example
neural
representations
in the
motor
system
of what
the arms
or hands
can
do

and
how
and
how
those
representations
are
engaged
when
you're
faced
with a
problem
like
picking up
a coffee
cup
or
getting a
nail out
of a
board
or the
hammer
your
system
this is
not
consciously
very
accessible
information
of course
kind of
knows how
to use
the
capabilities

of
the
limbs
to do
these
jobs
and
so
I think
a
question
for
the
discussion
that
Carl
was just
denunciating
about
attentional
control
and
metacognitive
representation
is this
question
of how
the
system
represents
its own
capabilities
from
how to
use
its
limbs
to

you
know
how to
manage
its
immune
system
sorts
of
things
that
Mark
Psalms
is
very
engaged
with
to
much
more
abstract
things
like
you know
do I
have the
cognitive
wherewithal
to read
this paper
and understand
it
we do
have some
sort of
understanding
of that
that's

difficult
to
articulate
so
I think
all of
those
questions
are
aspects
of
exactly
the
same
kind
of
question
of
internal
resource
representation
or
internal
capability
representation
which
an
attention
system
has to
have
if
it's
going
to
function
to
do

the
job
that
an
attention
system
has
to
do
so
I'm
afraid
I'm
going
to
have
to
depart
too
now
but
please
carry
on
as long
as you
like
and
thankfully
this is
being
recorded
so I
can
see
at
some
later

time
what
was
said
in
this
latter
part
so
thank
you
Carl
for
joining
and
thank
you
all
for
for
being
here
and
being
interested
thank
you
Chris
okay
does
anyone
have
a
comment
or
question
or
I

can
read
some
of
the
questions
in
the
live
chat
yes
Carl
well
yeah
I'm
just
picking
up
on
that
last
question
which
I
thought
was
extremely
interesting
I
noticed
you
put
the
strange
things
paper
up
on

the
chat
I
think
that
was
quite
astute
of
you
because
the
very
notion
of
an
inner
screen
tells
you
immediately
that
anything
lying
behind
that
inner
screen
does
not
have
direct
access
to
the
actuators
to
the

active
states
of
any
cell
or
any
person
or
indeed
any
population
and
that
has
quite
a
fundamental
implication
because
if
you
can
always
read
the
internal
states
within
a
Markov
blanket
!
or
at least
describe
them
as

making
inferences
about
the
cause
of
their
sensory
sector
what
that
means
is
that
the
internal
states
behind
the
inner
screen
will
also
be
representing
in
some
sense
in
some
minimal
sense
their
own
actions
because
the
actions

are
going
to
be
vicariously
a
cause
of
the
sensory
impressions
or
the
sensory
sector
of
the
Markov
blanket
that's
quite
remarkable
because
what
that
basically
means
is
that
if
you
have
an
inner
screen
you
effectively
are

inferring
what
you're
doing
you
have
Bayesian
beliefs
about
your
own
behavior
you're
not
directly
aware
of
in
the
sense
that
there's
a
dynamical
coupling
between
your
active
states
say
for
example
the
primary
afference
going
to
your

spinal
cord
so
the
spinal
cord
knows
exactly
the
consequences
of
its
action
but
the
central
nervous
system
or
parts
of
it
behind
the
inner
screen
do
not
so
you've
now
got
this
interesting
situation
where
you've
got

beliefs
posterior
beliefs
or
you can
be
described
as
having
as
if
you
had
posterior
beliefs
about
your
own
action
now
if
you've
got
posterior
beliefs
you
must
have
prior
beliefs
if
you've
got
prior
beliefs
about
your
own

action
before
it's
happened
you
have
intentions
so
at
some
level
there's
a
bright
line
between
things
that
do
and
do
not
have
intentions
and
I
would
just
coming
back
to
Chris
examples
of
the
base
phylogetic
trees

perhaps
even
simply
go
for
a
thermostat
a
thermostat
has
direct
access
to
the
consequences
of
its
action
whereas
I
do
not
so
you
could
argue
is there
a
different
kind
of
intentionality
or
does
does
a
thermostat
have

intentions
in
the
same
spirit
that
I
have
intentions
and
if
your
answer
is
yes
then
largely
you
are
answering
or you
are
making
that
discrimination
or
disambiguation
between
things
that
are
and
are
not
intentional
on
the
basis

of
possessing
an
inner
screen
and
then
you
can
have
a
further
argument
which
I
suspect
Concha
was
hinting
at
is
if
I'm
inferring
what
I'm
going
to
do
or
to
use
Chris's
words
what
am
I
doing

then
the
question
is
at
what
temporal
scale
is
the
doing
expressed
am
I
leading
a
good
life
am
I
going
to
the
shops
am
I
adjusting
my
posture
to
increase
my
comfort
am
I
reflexively
responding
to

heart
rate
variability
all of
these
are
very
different
time
constants
so I
think
there
is
a
question
now
of
the
temporal
depth
of
the
imagination
that
would
be
effectively
a
description
of
the
belief
updating
to
evaluate
to
infer

what
you
are
what
your
intentions
are
or
to
form
a
posterior
belief
that
will
then
be
enacted
by
your
reflexes
that
is
predicated
on
your
prior
beliefs
about
your
action
namely
your
intentions
so I
think
there's
possibly

a
vague
spectrum
of
things
that
just
look
very
myopically
into
the
future
say
viruses
and
things
that
look
quite
a
long
way
into
the
future
things
like
you
and
me
so I
think
there's
some
really
interesting
purely

technical
things
that
attend
any
belief
updating
or
any
system
that
is
equipped
with
an
inner
screen
that
can
then
be
described
as
if
it
is
doing
a
certain
kind
of
belief
updating
and
to
finish
the
the

recursion
and
the
metacognitive
aspect
inherits
from
the fact
that
you
don't
have
direct
access
to
the
consequences
of
your
actions
I'll
ask
a few
related
questions
here
in the
chat
so
Andreas
asks
can
we
introspect
our
Markov
blankets
or not

what
is
being
observed
who
or
what
is
there
to
observe
that
are
you
going
to
answer
that
Daniel
or
no
I
think
Chris
would
be
the
prime
person
to
answer
those
questions
I'm
not
sure
in this
question

of the
paper
when
there
is
a
Daniel
can you
just scroll
down
just to
make
sure
there's
not
a
figure
of
R
right
perhaps
that
was
it
I
think
there's
quite
a
straightforward
answer
to
that
question
just
looking
at
the

architecture
of the
kind
that's
been
presented
on
screen
now
so
if
there
is
any
hierarchical
or
heterarchical
structure
induced
by
the
sparse
coupling
of
any
dynamics
whether
they're
intracellular
dynamics
or whether
they're
the
connectome
in a
brain
then
that

hierarchical
structure
just
is
defined
by
a
series
of
nested
Markov
blankets
and just
to keep
things
simple
you can
imagine
a
sandwich
cake
or a
layer
cake
of
Markov
blanket
after
Markov
blanket
from the
point of
view of
any
given
Markov
blanket
what

is it
seeing
and
who
is it
talking
to
so
the
Markov
blanket
is
just
an
expression
of
the
inputs
and
the
outputs
so
in
this
simplified
picture
of
concentric
or
layer
cakes
hierarchical
subsubmission
hierarchy
of
Markov
blankets
then

effectively
one
level
is
basically
seeing
the
outputs
of
another
level
so
it's
basically
seeing
the
active
states
of
another
level
and
the
active
states
of
the
given
level
say
level
I
now
provide
the
inputs
namely
the

sensory
states
of
the
level
below
and
this
becomes
particularly
obvious
when you
think about
predictive
coding
hierarchies
so
predictive
coding
hierarchies
you have
this
notion
of
ascending
and
descending
messages
where
the
ascending
messages
are
effectively
the
free energy
gradients
or the

prediction
errors
usually
precision
weighted
prediction
errors
that
ascend
from
lower levels
of the
hierarchy
lower
Markov
blankets
if you
like
to
the
next
Markov
blanket
which
would
be
the
next
level
up
and
then
they
are
reciprocated
inducing
the
circular

causality
which
you
always
require
when
thinking
about
anything
embedded
in
its
environment
by
descending
predictions
so
the
actual
messages
what
is
seen
and
what
is
acted
well
in
this
instance
we're
associating
action
with
descending
predictions
but

usually
those
predictions
are going
to be
a
precision
to do
the
selection
we were
talking
about
but
more
generally
we can
think
of them
as
actions
on the
level
below
so
what that
basically
means
is that
if I'm
on
Markov
blanket
and I
have
hidden
states
behind

me
I am
basically
trying to
make
sense
of the
prediction
errors
that are
coming
from
the
Markov
blanket
below
me
and
in
making
sense
of
those
I
am
now
acting
on
those
blankets
by
issuing
predictions
that I
am now
going to
be using
in order

to
evaluate
the
prediction
error
and
then
if you
actually
write out
the
requisite
variables
in a
predictive
coding
scheme
and
just
look
at
the
links
you
can
you
can
quite
easily
identify
the
implicit
Markov
blankets
layer
after
layer
after

layer
layer
Do we have
more questions
for the
chat?
Do you
want to
ask
anything
or I
can
get another
one
chat
thing
I
have

question
from
the
figure
four
like
can
you
can
you
stay
there
on the
figure
four
like
every
reference
frame

has
in and
outside
connection
to the
executive
or
metacognitive
green
and
and

and
then
the
thermodynamic
free energy
distribution
quite a reference
frame
top one
has only an input
the output is
then
through valves
that
co-regulate
I guess
the quite a reference
frame activation
so yeah
I was thinking
why is there
only one
direction
and
isn't
then

the
thermodynamic
free energy
distribution
quantum reference
frame
that's a
terrible
name
by the
way
isn't
this
then
higher
in the
hierarchy
than
the
executive
which
could
be
answer
to the
question
that
dorsal
attention
overrides
like the
outer
nothing like
this
I'm not
sure
yeah
I think

a related
question
that I
have to
that
is like
if
layers
beget
another
layer
then
what
is
structurally
different
about
the
inner
screen
does
it
imply
similar
awarenesses
where
does
the
narrative
self
identification
relate
to
these
finite
hierarchies
which
come to a

stop just
through the
finiteness
of things
but that
might be a
Chris
question
related to
this
diagram
I guess
sort of
just in a
closing
ish
round
from
Adam
and
Carl
how do
you see
this
work
going
forward
what
would
you be
looking
to
reduce
your
uncertainty
with
further
what

kinds
of
predictions
or
experiments
do you
feel like
came out
of that
or
how did
the
discussions
of the
authors
relate
to
what
takeaways
and what
research
directions
arise
now
maybe
Adam
first
hmm
well
I never
know
actually
I'd
first
like to
be
surprised
in

terms
of
just
seeing
like
independently
what
people
come up
with
because
I
approach
this
from
a
very
different
take
than
being
grounded
in
the
formalism
I've
come
from
a
systems
neuroscience
and
cybernetic
perspective
but
and so
the ways
there are

differences
are in
some ways
most
interesting
to me
but
also
the
points
of
convergence
but
wondering
how far
we could
go
with
the
quasi
Cartesian
interpretations
like
not going
too far
but going
far
enough
wondering
whether
it can
bear
so like
earlier
Carl
mentioned
like
mental

transparency
but like
what
different
aspects
of
consciousness
studies
and
phenomenologies
can be
elucidated
by further
developments
like so
something I posted
in the
zoom chat
was like
there's been
suggestions
of
you know
there being
potentially
a Bayesian
blur problem
and
interpretations
of the
brain as a
general model
that could
produce
awareness
like why
doesn't
experience

seem
probabilistic
like why
don't we
get
superpositions
why does
this seem
so
classical
and so
like
Andy
Clark
for instance
discussed
this in
terms of
the requirement
of
consciousness
in terms
of its
functional
role
as being
coupled
to
mental
action
or
policy
selection
and
that doesn't
seem incompatible
with what's
being proposed

here
but I wonder
to what
degree
is this
inner screen
having
classical
properties
part of
what mediates
this
transition
or rather
part of
what would
explain
another way
not necessarily
mutually exclusive
because the
screen is
helping for
precision
weighting
and mental
action
and
intentional
selection
but to
what degree
is this
another
complementary
or separate
solution to
the so-called

Bayesian
blur problem
to the extent
that it is
a problem
and one more
thing I guess
I would wonder
is in terms
of like
unpacking
quasi-cartesian
intuitions
is so
like in the
work with
like Tim
like these
like two
data structures
they have
are like
a view
network
and a
pose
network
that are
like closely
coupled
for like
the state
of the
embodied
agent
relative to
the world
or for the

robot
and something
I would wonder
is like
this inner
screen
to what extent
does it have
like the
properties that
would correspond
to some of
the phenomenology
of space
like to what
degree is it
connected to
like visuospatial
awareness
but like to
what degree
does it
connect to
somatospatial
awareness
is that
subsumed
into visuospatial
awareness
is there
only one
modality
are there
multiple
spaces
like these
are I guess
some questions

I've wondered
about in terms
of like
what is the
relationship
between
like what
I guess
phenomenologists
might sometimes
describe as
the lived
body
and what
the folk
might call
or what people
might call
the mind's
eye
so I have
a lot
of different
questions
and wondering
whether further
developments
of this theory
could constrain
some of the
many places
I have
uncertainty
so just to
pick up on
some of the
key issues
that Adam's

just foregrounded
for us
so the
Bayesian
blur I only
understand through
Andy Clark's
writing and I
think Adam
picked up on a
very important
point there
that comes
back to what
we're talking
about before
in terms of
the differences
between intentions
and action
action is a
real variable
it's not a
random variable
you can only
infer a
random variable
so that
means that
your actual
action
including your
mental action
is not
a random
variable
and it cannot
have a

probability
distribution
it just
is
so the
manifestation
of your
mental
actions
can't have
a base
in blur
just in a
very simple
minded way
that can
also be read
in terms of
the act of
selection
you can only
select one
of many
things
and that's
why I quite
like it when
people use
the word
selection
or attention
selection
or selecting
this
as opposed
to that
and that
the act

of selection
is
you know
selection
is a verb
and it does
speak to
the
discrete
aspect
or the
unitary
aspect
of
manifesting
your
intentions
but cannot
have a
Bayesian
blur
so I think
that's an
interesting
thing
that inherits
from the
inner screen
the other
interesting
thing is
that
the inner
screen
because it
can no
longer
the innermost

screen
my apologies
because it
can no
longer be
sub-partitioned
thereby
enabling
more and
more abstract
explanations
for parts
of the
inner screen
the only
way that
it can
know
itself
is by
acting
on its
world
because
its world
is just
the rest
of the
brain
for us
or the
rest
of the
cell
for
a
bacteria
which

means
that it
can only
know
itself
through
acting
and that's
quite unique
to the
inner
screen
and what
does that
mean
in terms
of future
research
well it
means that
there may
be one
or more
I'm not
saying
there's
just one
inner
screen
there could
be a
number
of them
in a
heterarchical
structure
but it
does beg

the question
now what
is the
functional
anatomy
of this
inner
screen
from a
biological
point of
view
and that's
why I
deliberately
emphasised
the
neurobiology
and the
neuropharmacology
of this
mental
action
as
Mark
Soames
because we
know there
has to be
connections
from the
inner
screen
to the
cells of
origin
or the
things that

do the
mental
action
which
ultimately
reduce
to
classical
ascending
neuromodulatory
systems
so we've
got a
good start
in principle
for the
neurobiology
of it
on the
other hand
you might
start to
apply these
ideas and
the predictions
to
artificial
consciousness
so we wanted
to simulate
consciousness
or perhaps
we were
worried about
the ethics
of AI
alignment
and can

machines
suffer
are they
self-aware
and all
of these
questions
I think
rest very
much upon
the
functional
architectures
and the
implicit
information
geometries
that you
are
literally
in a
position
to
build
and then
you're
asking
questions
about
well
where
would
self-awareness
live
well it
would have
to live
in that

inner
screen
or in
that
innermost
screen
because
as
every
time
you
move
behind
these
successive
screens
you
are
from a
technical
point
of
view
compressing
your
fantasies
about
what's
causing
your
sensoria
of
sensoria
of
sensoria
and that
compression
ultimately

reduces
to the
simplest
explanation
for
my
sensorium
in that
it's
me
so
you
which
is just
another
hypothesis
but it's
a nice
simple
explanation
for
gathering
together
all these
different
modalities
that Adam
was talking
about
visual
modalities
the auditory
the
interoceptive
all of
these
can be
jointly

explained
by a
very simple
hypothesis
oh
I am
an
embodied
artifact
more
precisely
it's not
just
I am
one
because
if there
was
just
one
hypothesis
there
would
be
no
hypothesis
to
select
from
so
there
would
be
no
active
selection
so
you

would
never
know
that
but
as
soon
as
you
have
different
hypotheses
about
you
in
different
states
of
mind
or
you
in
different
states
of
being
you
can
now
act
upon
your
brain
and
ultimately
your
body
to

contextualize
and gather
evidence
yes
I am
in this
state
of
mind
everything
that I'm
receiving
from my
interceptive
modalities
visual
modalities
auditory
modalities
tells me
that
currently
I am
scared
I am
in a
dark
alley
in
a
city
that
I
do
not
know
my
heart

is
racing
I
feel
petrified
the best
explanation
for this
is that
it is
me
in
this
particular
state
of
mind
and
if
I
know
I'm
in
that
state
of
mind
then
I
can
act
by
selectively
attenuating
or
attending
to
various

sources
of
information
I
will
probably
in
this
instance
attend
acutely
to
my
auditory
modalities
less
to
my
visual
but
also
attend
acutely
or
in
fact
no
I
will
actually
attenuate
my
attention
to
my
bodily
states
in

preparation
for
flight
and
fight
and
of
course
in
so
doing
I
will
engage
different
kinds
of
reflexes
and
generate
more
evidence
that
I
am
in
a
state
of
panic
angst
or
at
least
arousal
and
alertness
that

would
prepare
me
so
to
come
back
to
where
I
started
would
you
want
to
build
that
into
a
robot
you
might
well
do
if
you
believe
people
like
Josje
Bach
for
example
if
you
want
to
elude

the
singularity
then
he
would
argue
you
really
need
to
make
our
artifacts
as
conscious
and
empathetic
as
we
are
so
you
might
have
to
move
in
that
sort
of
biometric
direction
in
order
to
guarantee
true
AI

alignment
right
across
the
board
a
great
place
to
take
it
and
leave
it
for
now
so
thank
you to
the
authors
for
the
very
provocative
paper
and for
joining
this
discussion
and to
Koya
for
precipitating
it
and giving
a lot of
thought to

it as
well
so best
luck
till
the
next
screen
stream
thanks
bye
bye
bye